>_ CYBER TECH ACADEMY

Linux Fundamentals

A practical textbook for navigating Linux, managing files, understanding permissions, and reading system activity.

LEVEL: BEGINNER

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How to use this guide

Read the concepts first, run examples only in a system you own or are authorised to use, complete the lab, then answer the review questions without looking back. Commands and outputs can differ by Linux distribution, software version, and local configuration.

Responsible learning: only test systems for which you have explicit permission. These notes are educational and do not grant permission to access, scan, disrupt, or modify third-party systems.

Learning objectives

- Explain what Linux, a distribution, a shell, and a terminal are.
- Navigate the filesystem and manipulate files safely.
- Read and change permissions using least privilege.
- Inspect processes, storage, memory, and text logs.
- Use pipes and redirection to build repeatable command workflows.

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1. Linux, distributions, the shell, and the filesystem

Linux is the kernel: the core component that manages hardware, memory, processes, devices, and system calls. A Linux distribution packages the kernel with utilities, libraries, a package manager, and often a desktop environment. Ubuntu, Debian, Fedora, Kali Linux, and Parrot OS are examples of distributions. Kali and Parrot include security tools, but they are not substitutes for learning the fundamentals.

The terminal is an interface for entering text commands. A shell such as Bash interprets those commands and starts programs. A command commonly contains a program name, options, and arguments. For example, in `ls -la /var/log`, `ls` is the program, `-la` are options, and `/var/log` is the argument.

```
pwd
ls -la
cd /var/log
pwd
cd ~
```

Linux paths start at `/`, the root directory. The home directory is usually represented by `~`. A leading slash means an absolute path; a path without one is normally interpreted relative to the current directory.

Safety habit: before using a command that changes or deletes data, check your current directory with pwd and inspect the target path. Avoid copying commands you do not understand into a privileged terminal.

2. Files, directories, and text

Use `mkdir` to create directories, `touch` to create an empty file, `cp` to copy, `mv` to move or rename, and `rm` to remove. The last command may permanently delete files, so test with a harmless temporary directory first.

```
mkdir -p ~/cyber-lab/notes
touch ~/cyber-lab/notes/lesson.txt
printf 'first lesson\nsecond lesson\n' > ~/cyber-lab/notes/lesson.txt
cat ~/cyber-lab/notes/lesson.txt
grep -n 'second' ~/cyber-lab/notes/lesson.txt
```

`cat` prints file contents; `less` allows page-by-page reading; `head` and `tail` show the beginning or end. `grep` searches for matching text, and `find` locates files by criteria.

```
find ~/cyber-lab -type f -name '*.txt'
file ~/cyber-lab/notes/lesson.txt
wc -l ~/cyber-lab/notes/lesson.txt
```

Quote paths containing spaces. Double quotes allow some variable expansion; single quotes preserve most characters literally. This matters when filenames or search terms contain shell metacharacters.

3. Users, groups, and permissions

Linux permissions control who can read, write, or execute a file. The three permission classes are owner, group, and others. In `-rwxr-x---`, the owner has read, write, and execute; the group has read and execute; others have no permissions. For directories, execute permission means the ability to traverse/search the directory.

```
id
ls -l ~/cyber-lab/notes/lesson.txt
chmod 600 ~/cyber-lab/notes/lesson.txt
ls -l ~/cyber-lab/notes/lesson.txt
```

Numeric permissions add values: read = 4, write = 2, execute = 1. Thus 7 means read/write/execute, 6 means read/write, and 5 means read/execute. `chmod 600 file` gives the owner read/write access and removes access for group and others.

`sudo` runs a command with elevated privileges when permitted. Use it only when necessary. A mistaken privileged command can alter the whole system. Do not change ownership or permissions on system directories just to silence an error.

Least privilege means granting only the access required for a task, for only as long as needed. It reduces accidental damage and limits the impact of a compromised account.

4. Processes and system health

A process is a running program with a process ID (PID). Tools such as `ps` and `top` show process activity. Use `free -h` for memory, `df -h` for filesystem capacity, and `du -sh directory` for a directory's approximate size.

```
ps aux | head
free -h
df -h
uptime
```

Stopping a process should be deliberate. `kill PID` sends SIGTERM by default, asking a process to stop gracefully. A forceful signal such as SIGKILL prevents cleanup and should not be the first choice. Confirm the PID before acting; never guess.

System utilities vary across distributions. If a command is missing, consult the manual or the distribution's package documentation rather than downloading an unknown binary.

5. Pipes, redirection, and logs

A pipe (`|`) sends one program's standard output to another program's standard input. Redirection sends output to a file: > overwrites a file and >> appends. Standard error is a separate stream, useful for distinguishing diagnostic messages from normal output.

```
printf 'INFO startup\nERROR disk full\nINFO retry\n' > ~/cyber-lab/sample.log
grep -i 'error' ~/cyber-lab/sample.log
grep -ic 'error' ~/cyber-lab/sample.log
```

Log files help administrators understand system behaviour. A log entry often contains a timestamp, component, severity, and message. Interpret entries in context: one error may be harmless, while repeated failures correlated with other events may indicate a larger problem.

Shell scripts automate repeated tasks. Begin with small scripts, quote variables, validate inputs, and use a test directory. Never run a script from an unknown source with elevated privileges.

Guided practical lab

Goal: create a small workspace, protect a file, and analyse a log without changing system files.

- Create ~/cyber-lab/notes and a file called checklist.txt.
- Write three lines of fictional system events to sample.log.
- Use grep to find all lines containing WARN or ERROR; count the errors.
- Set checklist.txt to owner-only read/write permissions and verify with ls -l.
- Use df -h and free -h to record disk and memory information.

```
mkdir -p ~/cyber-lab/notes
printf 'INFO service started\nWARN retrying connection\nERROR sample failure\n' > ~/cyber-lab/sample.log
grep -E 'WARN|ERROR' ~/cyber-lab/sample.log
grep -c 'ERROR' ~/cyber-lab/sample.log
chmod 600 ~/cyber-lab/notes/checklist.txt
ls -l ~/cyber-lab/notes/checklist.txt
```

Expected result: grep shows the warning and error lines; the error count is 1; permissions show read/write for the owner and no permissions for group or others. Use only your own test files.

Knowledge check

Try to answer these before checking the answer key. Write your reasoning, not just a single word.

1. What is the difference between the Linux kernel and a Linux distribution?

Answer: _____

2. What does chmod 640 file.txt mean?

Answer: _____

3. What is the difference between > and >>?

Answer: _____

4. Why should sudo be used carefully?

Answer: _____

5. What does grep -i do?

Answer: _____

Answer key and reasoning

1. The kernel manages core system resources; a distribution packages the kernel with utilities, libraries, package management, and often a desktop environment.

2. Owner can read and write (6); group can read (4); others have no permissions (0).

3. Both redirect output. > overwrites or creates the destination file; >> appends to it.

4. Elevated privileges can change system-wide files and settings; mistakes or malicious commands can cause broad damage.

5. It performs a case-insensitive text search.

Glossary

Term	Meaning
Kernel	Core software that manages hardware and system resources.
Shell	Program that interprets commands.
PID	Unique process identifier.
Permission	Rule controlling access to a file or directory.
Pipe	Connects one command's output to another command's input.
Least privilege	Granting only the access needed for a task.

Further study

Continue with the next Cyber Tech Academy module, practise in a disposable virtual machine or an explicitly authorised lab, and consult official manuals for the tools you use. Verify commands against your own system before running them.

Cyber Tech Academy • Free learning • Learn the theory. Practise safely. Build real skills.